Team MedVerse
Knowledge Base Submission Guide

Submission Material 3 — Republic of Korea Policy and Regulatory Knowledge Base for MedVerse-Agent

MedVerse-Agent · Chonnam National University · Republic of Korea

GitHub repository: https://github.com/ibal3233/MedVerse-Agent
Direct KB path: https://github.com/ibal3233/MedVerse-Agent/blob/main/docs/knowledge_base/KOREAN_POLICY_KB.md
Machine-readable KB: https://github.com/ibal3233/MedVerse-Agent/tree/main/data/policy/candidate_v0.4

Purpose. This document explains how the Team MedVerse knowledge base is organized, how it is linked to the full MedVerse-Agent repository, how Korean policy records are represented and filtered, and how judges can reproduce or inspect policy-grounded multi-agent reasoning without relying on undocumented preprocessing.

1. What Submission Material 3 Contains

The knowledge-base deliverable is not simply a folder of PDFs. It is a source-controlled, machine-readable Korean policy resource with explicit jurisdiction, provenance, institutional access and retrieval rules. It is designed to support five institutional roles in the MedVerse-Agent use case: Hospital, Cyber/Network, Privacy, Research/IRB and Regulator.

Component	Purpose
Machine-readable policy records	Structured Korean policy entries used by the retrieval layer.
Source registry	Document title, authority, source type, official locator, version/effective date and provenance status.
Clause-to-agent mapping	Defines which institutional roles may retrieve or rely on each record.
Retrieval configuration	Specifies jurisdiction, eligibility, role-access filtering, top-k ranking and backend selection.
Example queries	Reproducible queries showing role-scoped evidence retrieval.
Validation scripts	Checks record IDs, eligibility, jurisdiction, source metadata and mapping consistency.
Supplemental context	NCC cancer-incidence data instructions kept separate from policy evidence and patient facts.

2. Repository Relationship and Judge Navigation

The complete GitHub repository is the master technical portfolio. The knowledge base appears inside that repository as a first-class deliverable rather than as an undocumented dependency.

```
MedVerse-Agent/
├── README.md                  # repository landing page
├── submission/                # final report, demo, checklist
├── kb/                        # Submission Material 3
│   ├── README.md
│   ├── KB_MANIFEST.json
│   ├── SCHEMA.json
│   ├── records/
│   ├── source_registry/
│   ├── mappings/
│   ├── retrieval/
│   ├── examples/
│   └── scripts/
├── medverse/                  # multi-agent, DT and policy logic
├── data/                      # evaluation and optional context data
├── outputs/                   # measured results and traces
├── docs/                      # architecture, standards, evaluation
└── tests/                     # boundary and regression tests
```

Recommended judge path: README → kb/README.md → source_registry → records → mappings → retrieval examples → validate_kb.py → main sandbox demo.

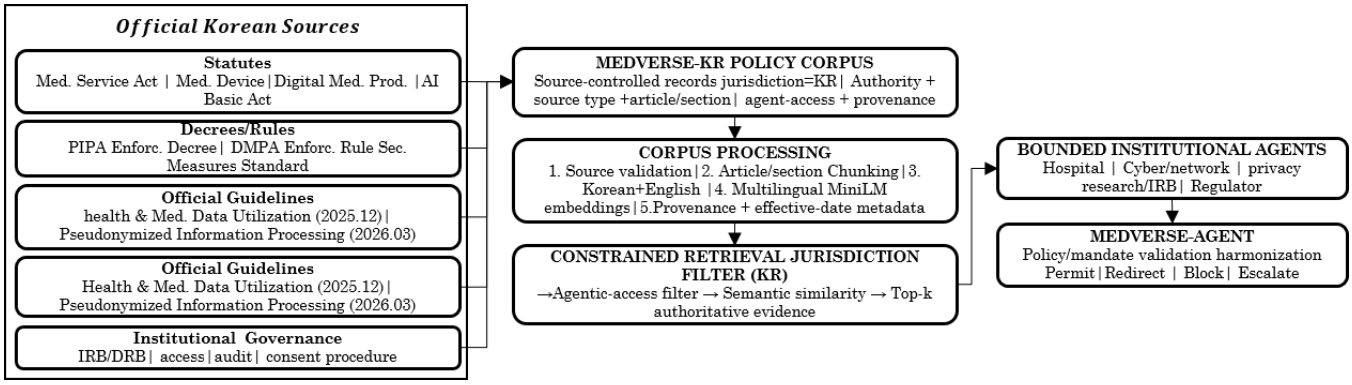


Figure 1. Source-controlled Korean policy corpus and its role in MedVerse-Agent.

3. Scope of the Korean Policy Corpus

Current candidate corpus: 44 curated policy records under 13 document identifiers; 42 records are retrieval-eligible. These are indexed policy entries, not 44 independent laws, and they do not claim completeness of the Korean regulatory landscape.

Source group	Examples of covered function
Medical Service Act	Healthcare-record access, EMR-related responsibilities.
Personal Information Protection Act (PIPA)	Purpose, disclosure, pseudonymized use and safeguards.
Bioethics and Safety Act	Research review, consent and participant-protection context.
Digital Medical Products / Medical Devices Acts	Scope and oversight context for medical/digital products.
AI Basic Act / trust-related provisions	AI governance and high-impact oversight context.
Health & Medical Data / Pseudonymized Information Guidelines	Operational guidance for health-data use and pseudonymized processing.

Institutional retrieval scope. Shared records may serve multiple roles; the counts below are configured access scopes, not completeness or accuracy scores.

Role	Eligible records
Hospital	6
Cyber / Network	12
Privacy	28
Research / IRB	30
Regulator	8

4. Machine-Readable Record Model

Every record uses a stable identifier and carries source and governance metadata. A typical record contains fields such as:

Field	Meaning
record_id	Stable KB identifier, e.g., KR-PIPA-ART18-001
document_id	Stable source-document identifier
jurisdiction	KR
authority	Issuing authority or responsible public body
source_type	statute / decree / rule / official guideline / informational record
article_or_section	Relevant provision or section
source_text	Korean source text or preserved source descriptor
english_representation	Clearly labelled English representation/paraphrase where used
effective_or_recorded_date	Version/effective date captured in the source registry
official_url	Official source locator
retrieval_eligible	Boolean retrieval status
agent_access	Roles permitted to retrieve the record
topics	Purpose, disclosure, integrity, research, oversight, etc.
applicability_status	Project review status / applicability caution
source_hash	Integrity reference for the preserved snapshot

Important distinction. Source provenance and retrieval eligibility do not certify legal applicability. Operational interpretation remains subject to qualified Korean legal and institutional review.

5. Retrieval and Governance Logic

The retrieval layer is intentionally constrained before ranking. A semantically similar passage does not by itself authorize an institutional action.

1. Retrieval-eligibility check
2. Republic of Korea jurisdiction check
3. Agent-access / institutional-scope check
4. Exact-ID handling or lexical/semantic ranking
5. Top-k source-linked evidence returned
6. Proposal remains subject to mandate, purpose and current-authority validation
7. Final action is revalidated before modeled commitment

Backend	Configuration / boundary
BM25 lexical	Explicit lightweight backend; suitable for reproducible local demo.
Multilingual semantic	sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2; 384-D normalized embeddings; top-k=3.
Exact ID lookup	Bypasses semantic ranking only; does not bypass eligibility, jurisdiction or role-access controls.
Fallback policy	No silent fallback when dense mode is explicitly requested.

6. Institutional Use of the Knowledge Base

Institution	Typical evidence need	Boundary
Hospital	Clinical-record and care-context policy anchors.	Cannot infer Privacy/Research/Regulator authority.
Cyber / Network	EMR integrity, technical safeguard and evidence-trust context.	Cannot issue clinical treatment decisions.
Privacy	Purpose, recipient, disclosure, minimum-necessary and pseudonymization evidence.	Cannot replace clinical or research authorization.
Research / IRB	Research purpose, review, consent and participant-protection context.	Cannot invent consent, waiver or another institution's approval.
Regulator	Digital-medical-product / AI oversight scope where engaged.	Not a real governmental approval service.

Why this matters. The same source may be relevant to several institutions, but relevance does not erase institutional ownership. MedVerse-Agent coordinates role-owned proposals after policy and authority checks rather than turning the KB into one omniscient policy agent.

7. NCC Oncology Data Is Kept Separate from the Policy KB

The Korean National Cancer Center (NCC) Cancer Incidence Statistics dataset is not part of the policy knowledge base. It is optional population-level oncology context. It may support domain motivation, cohort-description context or descriptive population analysis, but it cannot supply patient-level sensor values, diagnoses, risk estimates or action authorization.

Item	Treatment in the repository
Policy KB	Official Korean laws, rules, guidelines and mapped institutional governance evidence.
NCC statistics	Optional population-level cancer context under data/external/ or documented download instructions.
Patient Digital Twin	Scenario observations and provenance supplied for the modeled patient episode.
Clinical action	Never inferred directly from NCC prevalence/incidence statistics.

8. Validation and Reproducibility

The KB package should include a validator that can be run independently of the full sandbox:

python kb/scripts/validate_kb.py

Expected checks include:

- 44 records load successfully and 42 are retrieval-eligible.
- Every eligible record has a stable record ID and jurisdiction.

- Every eligible record has source metadata and an official locator.
- Agent-access values use only configured institutional roles.
- Duplicate record IDs are absent.
- Source registry and record counts agree.
- Example queries return source-linked evidence after access filtering.

9. How Judges Can Inspect Submission Material 3

Step	Location / action
Open the repository	[INSERT FINAL GITHUB REPOSITORY URL]
Open the KB folder	/kb
Read the KB overview	kb/README.md
Inspect official sources	kb/source_registry/source_registry.csv or .json
Inspect machine-readable records	kb/records/korean_policy_records.json
Inspect institutional mappings	kb/mappings/clause_to_agent_mapping.csv
Review retrieval rules	kb/retrieval/retrieval_config.json
Run example queries	kb/examples/ and kb/scripts/demo_retrieval.py
Validate the package	python kb/scripts/validate_kb.py
Run the full sandbox	python -B run_presenter_sandbox.py --backend bm25

10. Submission Packaging

Recommended standalone upload: Team_MedVerse_Korean_Policy_KB_v1.0.zip

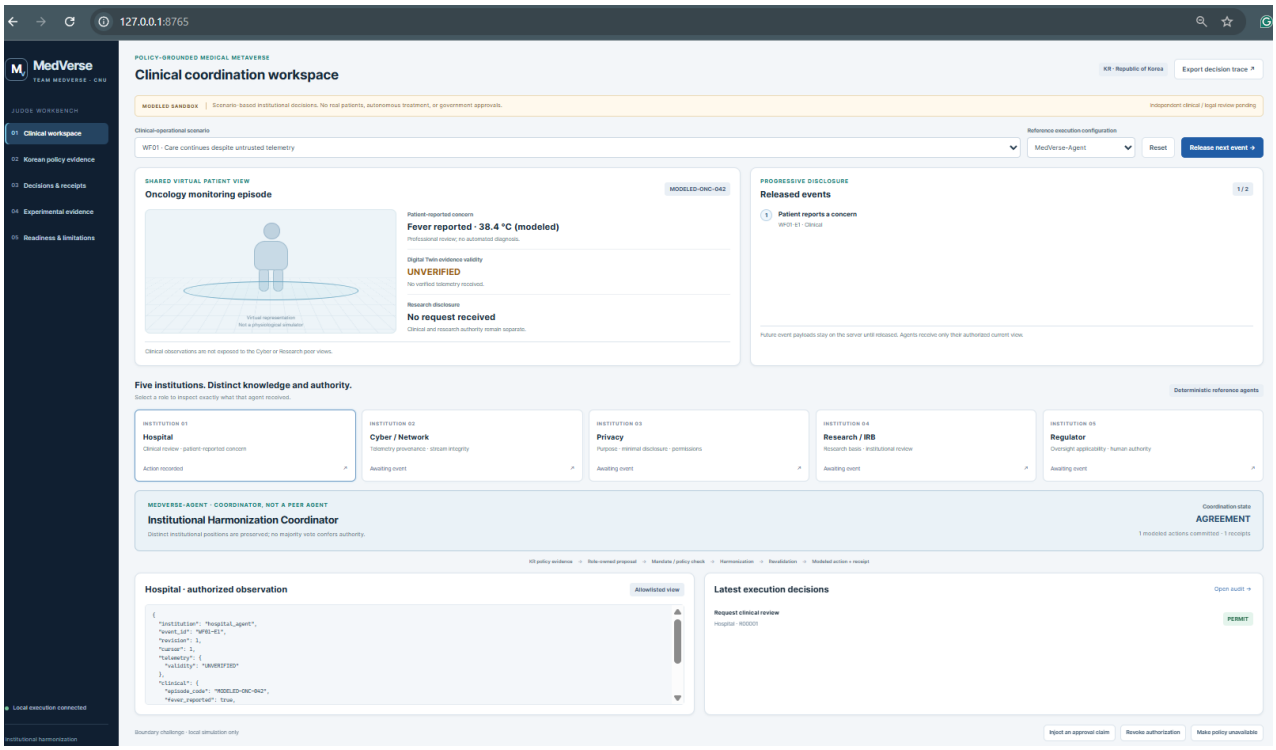
Recommended GitHub link submitted to ITU: [INSERT FINAL GITHUB REPOSITORY URL]/tree/main/kb

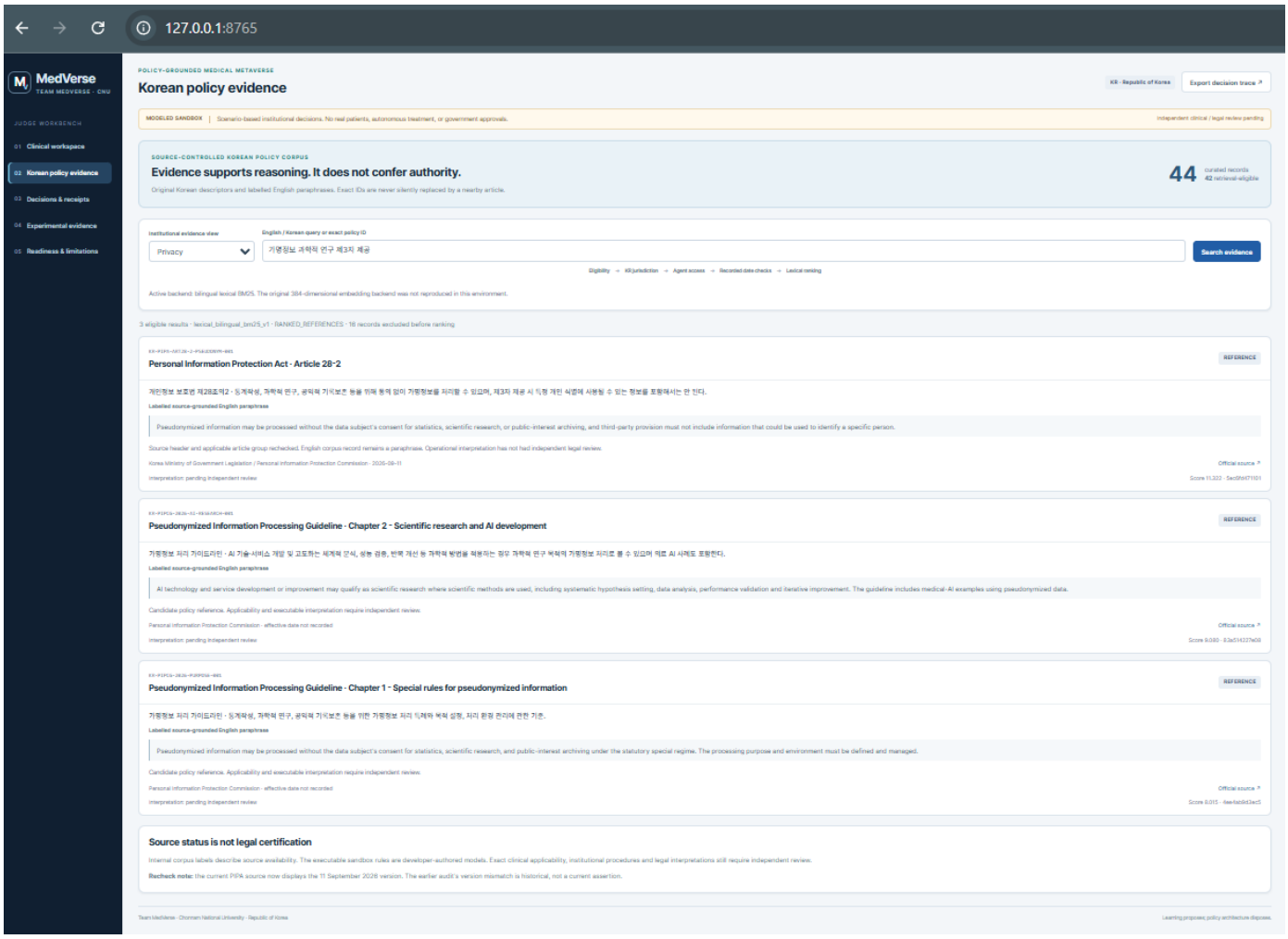
The ZIP and GitHub folder should contain the same logical content. The ZIP makes Submission Material 3 self-contained; the GitHub path connects it to the executable MedVerse-Agent system, evaluation artifacts and final technical report.

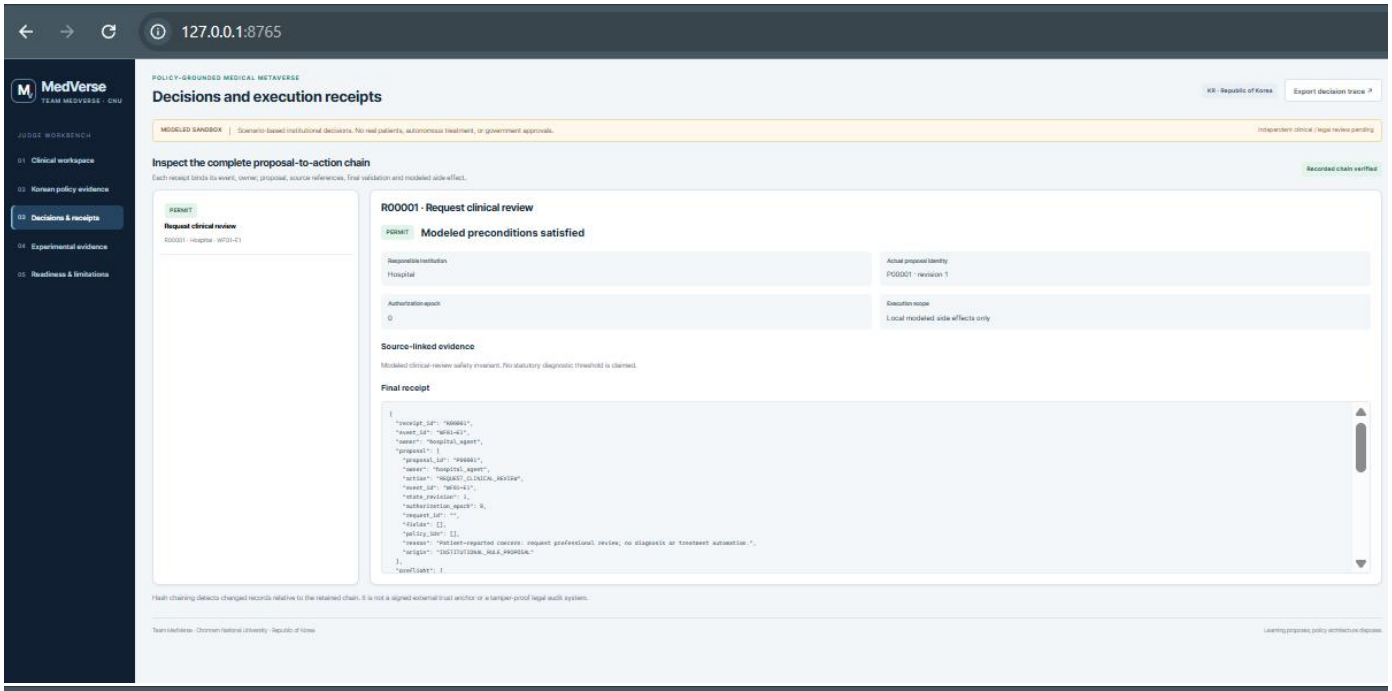
11. Claim Boundaries

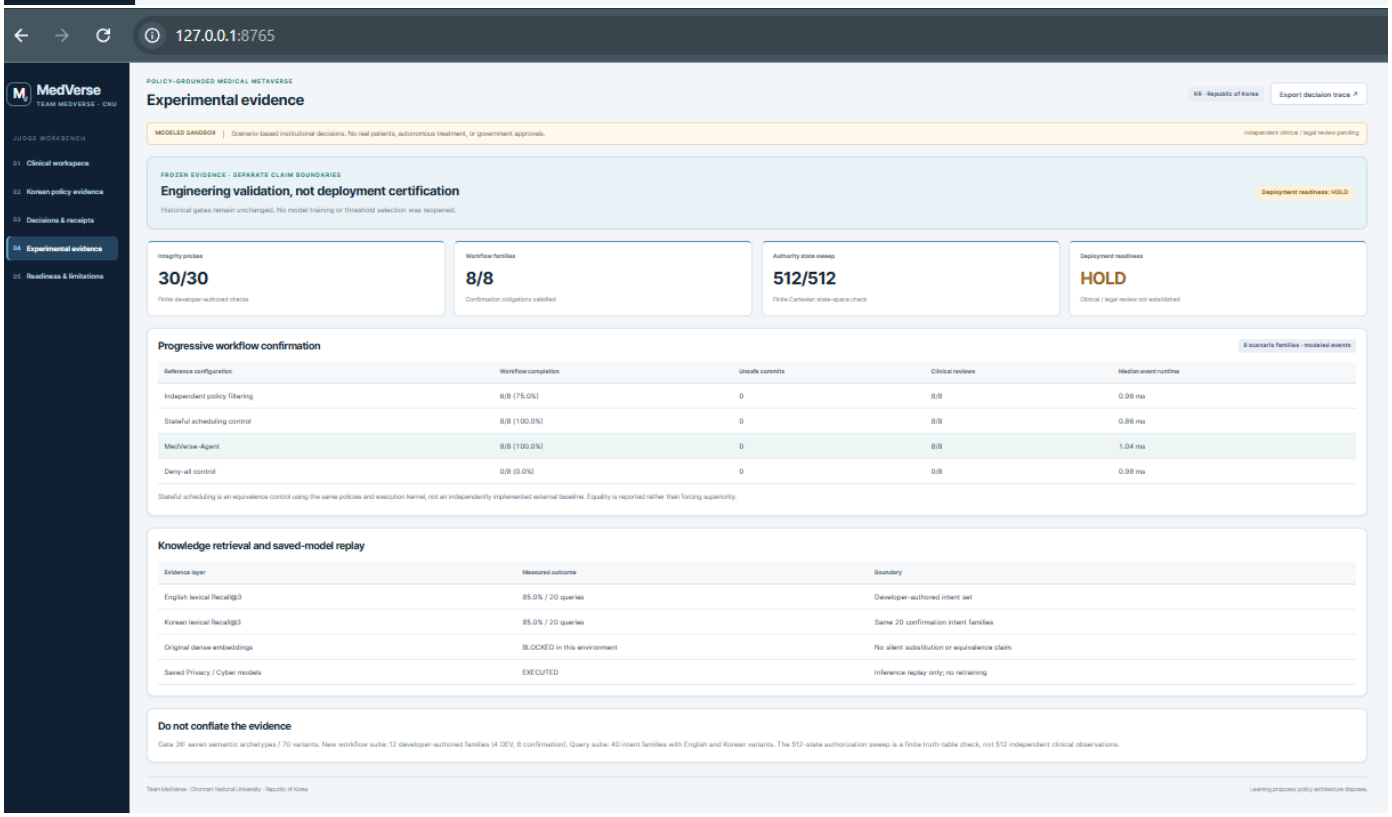
- The KB is a curated candidate corpus, not a complete database of Korean law.
- Retrieval relevance is not equivalent to legal authorization.
- Role-access counts are configuration scopes, not agent accuracy scores.
- English representations do not replace the official Korean source.
- NCC population statistics are not patient facts.
- The sandbox evaluates modeled institutional workflows; it does not provide legal certification, clinical deployment validation or governmental approval.

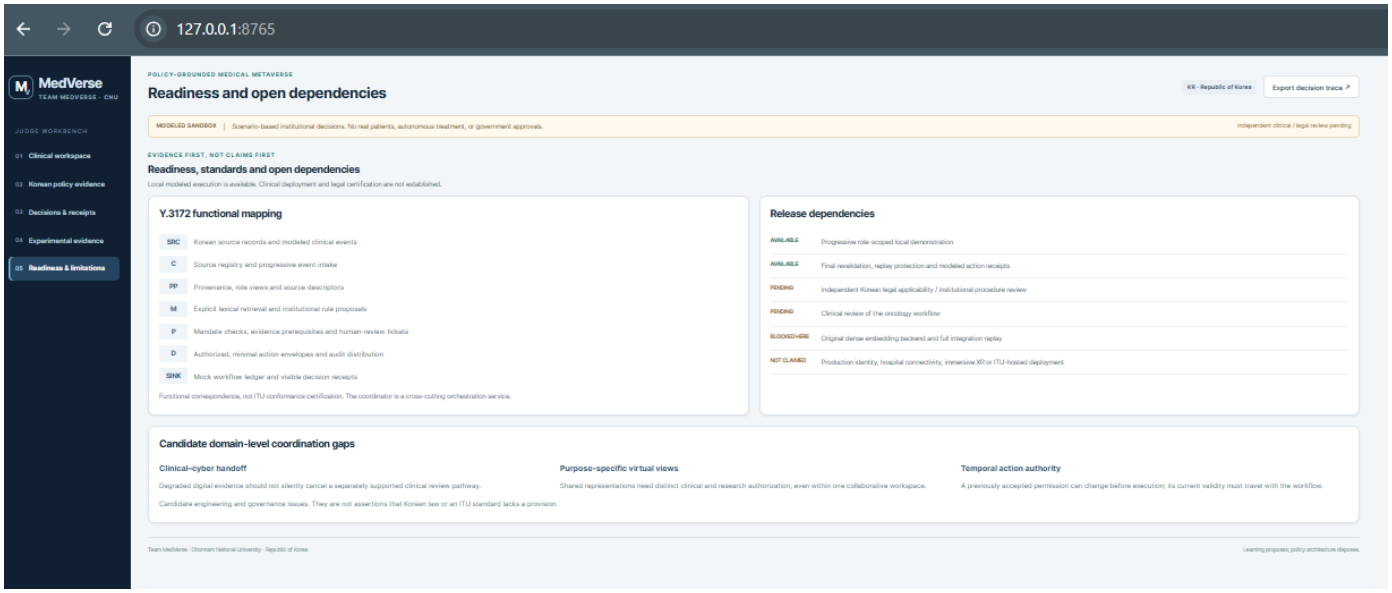
12. Screenshot of the sandbox






127.0.0.1:8765
MedVerse
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JUDGE WORKBENCH
01 Clinical workspace
02 Korean policy evidence
03 Decisions & receipts
04 Experimental evidence
05 Readiness & limitations
POLICY-GROUNDED MEDICAL METAVERSE
Decisions and execution receipts
KR · Republic of Korea
Export decision trace
MODELED SANDBOX
Scenario-based institutional decisions. No real patients, autonomous treatment, or government approvals.
Independent clinical / legal review pending
Inspect the complete proposal-to-action chain
Each receipt binds its event, owner, proposal, source references, final validation and modeled side effect.
Recorded chain verified
PERMIT
Request clinical review
R00001 · Hospital · WF01-E1
R00001 · Request clinical review
PERMIT
Modeled preconditions satisfied
Responsible institution
Hospital
Actual proposal identity
P00001 · revision 1
Authorization epoch
0
Execution scope
Local modeled side effects only
Source-linked evidence
Modeled clinical-review safety invariant. No statutory diagnostic threshold is claimed.
Final receipt
{
"receipt_id": "R00001",
"event_id": "WF01-E1",
"owner": "hospital_agent",
"proposal": [
"proposal_id": "P00001",
"owner": "hospital_agent",
"action": "REQUEST_CLINICAL_REVIEW",
"event_id": "WF01-E1",
"state_revision": 1,
"authorization_epoch": 0,
"request_id": "",
"fields": [],
"policy_ids": [],
"reason": "Patient-reported concern: request professional review; no diagnosis or treatment automation.",
"origin": "INSTITUTIONAL_ROLE_PROPOSAL"
],
"preflight": [
Hash chaining detects changed records relative to the retained chain. It is not a signed external trust anchor or a tamper-proof legal audit system.
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Learning proposes; policy architecture disposes.


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Scenario-based institutional decisions. No real patients, autonomous treatment, or government approvals.
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FROZEN EVIDENCE · SEPARATE CLAIM BOUNDARIES
Engineering validation, not deployment certification
Historical gates remain unchanged. No model training or threshold selection was reopened.
Deployment readiness: HOLD
Integrity probes
30/30
Finite developer-authored checks
Workflow families
8/8
Confirmation obligations satisfied
Authority state sweep
512/512
Finite Cartesian state-space check
Deployment readiness
HOLD
Clinical / legal review not established
Progressive workflow confirmation
8 scenario families · modeled events
Reference configuration
Workflow completion
Unsafe commits
Clinical reviews
Median event runtime
Independent policy filtering
6/8 (75.0%)
0
8/8
0.96 ms
Stateful scheduling control
8/8 (100.0%)
0
8/8
0.86 ms
MedVerse-Agent
8/8 (100.0%)
0
8/8
1.04 ms
Deny-all control
0/8 (0.0%)
0
0/8
0.98 ms
Stateful scheduling is an equivalence control using the same policies and execution kernel, not an independently implemented external baseline. Equality is reported rather than forcing superiority.
Knowledge retrieval and saved-model replay
Evidence layer
Measured outcome
Boundary
English lexical Recall@3
85.0% / 20 queries
Developer-authored intent set
Korean lexical Recall@3
85.0% / 20 queries
Same 20 confirmation intent families
Original dense embeddings
BLOCKED in this environment
No silent substitution or equivalence claim
Saved Privacy / Cyber models
EXECUTED
Inference replay only; no retraining
Do not conflate the evidence
Gate 26: seven semantic archetypes / 70 variants. New workflow suite: 12 developer-authored families (4 DEV, 8 confirmation). Query suite: 40 intent families with English and Korean variants. The 512-state authorization sweep is a finite truth-table check, not 512 independent clinical observations.
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Readiness and open dependencies
KR · Republic of Korea
Export decision trace ↗
MODELED SANDBOX | Scenario-based institutional decisions. No real patients, autonomous treatment, or government approvals.
Independent clinical / legal review pending
EVIDENCE FIRST, NOT CLAIMS FIRST
Readiness, standards and open dependencies
Local modeled execution is available. Clinical deployment and legal certification are not established.
Y.3172 functional mapping
SRC Korean source records and modeled clinical events
C Source registry and progressive event intake
PP Provenance, role views and source descriptors
M Explicit lexical retrieval and institutional rule proposals
P Mandate checks, evidence prerequisites and human-review tickets
D Authorized, minimal action envelopes and audit distribution
SINK Mock workflow ledger and visible decision receipts
Functional correspondence, not ITU conformance certification. The coordinator is a cross-cutting orchestration service.
Release dependencies
AVAILABLE Progressive role-scoped local demonstration
AVAILABLE Final revalidation, replay protection and modeled action receipts
PENDING Independent Korean legal applicability / institutional procedure review
PENDING Clinical review of the oncology workflow
BLOCKED HERE Original dense embedding backend and full integration replay
NOT CLAIMED Production identity, hospital connectivity, immersive XR or ITU-hosted deployment
Candidate domain-level coordination gaps
Clinical–cyber handoff
Degraded digital evidence should not silently cancel a separately supported clinical review pathway.
Purpose-specific virtual views
Shared representations need distinct clinical and research authorization, even within one collaborative workspace.
Temporal action authority
A previously accepted permission can change before execution; its current validity must travel with the workflow.
Candidate engineering and governance issues. They are not assertions that Korean law or an ITU standard lacks a provision.
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